Geometry
Unit 9
Lesson 11 Homework

Name___Answer Key________
Date______________________
Period_____________________

For questions 1-10, use $\pi = 3.14$ and round all answers to the nearest hundredth if necessary.

1. When Felix was young, he had a mini trampoline. It was 36 inches in diameter. What was the base (jumping) area of the mini trampoline?

   $r = 18$
   $A = \pi(18)^2 = 324\pi \approx 1017.88$ in^2 (when using the pi button)

2. As Felix grew bigger, he needed a bigger trampoline. Felix got a new trampoline for his birthday. It was 128 inches in diameter. What was the base (jumping) area of the new trampoline?

   $r = 64$
   $A = \pi(64)^2 = 4096\pi \approx 12{,}867.96$ in^2 (when using the pi button)

3. What is the difference in areas of the two trampolines?

   $12867.96 - 1017.88 = 11{,}850.08$ in^2

4. With the frame, Felix's new trampoline stood 35 inches high. Felix wants to put a skirt around the trampoline to prevent his pets from going underneath it while he is jumping. What is the lateral surface area of the new trampoline that is to be covered by the skirt?

   $2\pi(64)(35) \approx 14{,}074.34$ in^2 (when using the pi button)

5. Although his pets could not fit under the mini trampoline, what was the lateral surface area of the mini trampoline if it stood 13 inches high?

   $2\pi(18)(13) \approx 1470.27$ in^2 (when using the pi button)

6. Hermione is decorating for a birthday party. She has four sizes of gift boxes that are to be placed in one corner, simply for decoration. The base of one is 8 inches by 12 inches, the second is 12 inches by 16 inches, the third is 18 inches by 24 inches, and the fourth is 20 inches by 20 inches. What is the base area of the smallest box?
$8 \times 12 = 96$ in^2

7. What is the base area of the second box?
$12 \times 16 = 192$ in^2

8. What is the base area of the third box?
$18 \times 24 = 432$ in^2

9. What is the base area of the largest box?
$20 \times 20 = 400$ in^2

10. Hermione designated space in the corner for the gift boxes that is 50 inches by 60 inches. Can she fit all 4 sizes of gift boxes in the designated space?
$96 + 192 + 432 + 400 = 1120$
$1120 < 3000$
Yes, the total area available is 3000 in^2 and the boxes only take up a total of 1120 in^2.